

SIWES Cybersecurity Lab – Week 2 Report

Student Name: Hagos Abdulalim Jemal

Program: Cybersecurity

Repository Name: SIWES-CYBERSECURITY-LAB

Date: September 18, 2026

1. Executive Summary

During Week 2, I configured local Linux environments and performed system identification, file management, access control configuration, process and service auditing, environment variable inspection, and system log text searching. All outputs were captured as evidence and pushed to the remote GitHub repository.

2. System Identification & File Management

Task 1: System Identification

- **Commands Executed:** `uname -a, hostnamectl, cat /etc/os-release`
- **Purpose:** Retrieved kernel version, system hostname, and Linux distribution details (Kali Linux).
- **Evidence File:** `Week_02/06_Evidence/W2_01_System_Info.png`

Task 2: File Management & Navigation

- **Commands Executed:** `mkdir, touch, cp, mv, rm`
- **Purpose:** Navigated directory trees, created file structures, copied/moved artifacts, and safely deleted test files.
- **Evidence File:** `Week_02/06_Evidence/W2_02_Files.png`

3. Security & Operations Inspection

Task 3: Users, Groups, and Permissions

- **Commands Executed:** `whoami, id, ls -l, chmod, chown`
- **Key Findings:** Identified current user group memberships and modified Linux file permissions using numeric (`chmod 755`) and symbolic modes to enforce principle of least privilege.
- **Evidence File:** `Week_02/06_Evidence/W2_03_Permissions.png`

Task 4: Processes Inspection

- **Commands Executed:** `ps, ps aux, top`
- **Evidence File:** `Week_02/06_Evidence/W2_04_Processes(1).png`,
`Week_02/06_Evidence/W2_04_Processes(2).png`

Process Name	Owner	Purpose
systemd	root	Primary initialization process (PID 1) responsible for booting and managing system services.
bash	user	Interfacing command-line shell parsing user terminal operations.
NetworkManager	root	Network interface detection and connection configuration daemon.
dbus-daemon	messagebus	Inter-process communication bus enabling desktop services to communicate.
sshd	root	Listens for incoming encrypted SSH connections.

4. Service Audit & Environment Variables

Task 5: Service Investigation

- **Command Executed:** `systemctl --type=service`
- **Evidence File:** `Week_02/06_Evidence/W2_05_Services(1).png`,
`Week_02/06_Evidence/W2_05_Services(2).png`
- **accounts-daemon.service:** Manages local user account details via D-Bus. Operates strictly locally; does **not** communicate over external network interfaces.

- **ufw.service:** Uncomplicated Firewall service controlling system netfilter rules. Operates directly on network traffic (Network-Filtering).
- **ssh.service:** Provides secure remote command-line access. Communicates over network.

Task 6: Environment Variables

- **Commands Executed:** `echo $USER, echo $HOME, echo $SHELL, echo $PATH`
- **Evidence File:** Week_02/06_Evidence/W2_06_Env_Vars.png
- **USER:** Defines active account identity.
- **HOME:** Base path for personal user configuration and files.
- **SHELL:** Default CLI interpreter executable (`/bin/bash` or `/bin/zsh`).
- **PATH:** Ordered directory list searched by Linux to execute entered commands.

5. Filesystem Architecture & Log Analysis

Task 7: Filesystem Exploration

- **Commands Executed:** `ls -ld /home /etc /var /var/log /tmp /usr /proc, ls -la /var/log`
- **Evidence File:** Week_02/06_Evidence/W2_07_Fileystem(1).png, Week_02/06_Evidence/W2_07_Fileystem(2).png
- **/etc:** System-wide administrative configurations.
- **/var/log:** Centralized storage for application, security, and OS event logs.
- **/tmp:** Transient temporary storage cleared on reboot.
- **/proc:** Virtual dynamic directory displaying real-time kernel memory/process state.

Task 8: Log Inspection & Text Searching

- **Commands Executed:** `head -n 10, tail -n 10, grep -i`
- **Evidence File:** Week_02/06_Evidence/W2_08_Log_Analysis.png
- **Analysis:** Demonstrated non-destructive log inspection. Used `head` to sample log headers, `tail` to observe recent system events, and `grep` to filter target security strings without opening full log files into memory.

6. Verification Summary

All eight evidence files have been verified, staged, committed, and synced to the remote GitHub repository under `Week_02/06_Evidence/`:

- `W2_01_System_Info.png`
- `W2_02_Files.png`
- `W2_03_Permissions.png`
- `W2_04_Processes(1).png`
- `W2_04_Processes(2).png`
- `W2_05_Services(1).png`
- `W2_05_Services(2).png`
- `W2_06_Env_Vars.png`
- `W2_07_Filessystem(1).png`
- `W2_07_Filessystem(2).png`
- `W2_08_LogAnalysis.png`